BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 8475 / 8472

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8475

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. At high frequencies, a R-L-C series circuit behaves like:

(A) R-C circuit
(B) R-L circuit
(C) Capacitive circuit
(D) Parallel resonance circuit

Q2. At what frequency will an inductor of 1H have a reactance of 500Ω ?

(A) 500Hz
(B) 250Hz
(C) 350Hz
(D) 80Hz

Q3. The back motor effect in generators is in agreement with the law of conservation of:

(A) Charge
(B) Momentum
(C) Energy
(D) Mass

Q4. The turns ratio Ns/Np of a step-up transformer is:

(A) Greater than 1
(B) Less than 1
(C) Equal to 1
(D) Equal to zero

Q5. Lorentz force is given by:

(A) $\vec{F} = q(\vec{v} \times \vec{B})$ (B) $\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$

(C) $\vec{F} = q\vec{E}$ (D) $\vec{F} = I(\vec{L} \times \vec{B})$

Q6. The SI unit of permeability of free space is:

(A) Wb (B) Wbm^{-2}

(C) $WbA^{-1}m^{-1}$ (D) It has no unit

Q7. The charge carriers in electrolytes are:

(A) Free electrons (B) Holes

(C) Protons (D) Positive and negative ions

Q8. If time constant of R-C circuit is small, the capacitor is charged or discharged:

(A) Rapidly (B) Slowly

(C) Intermediately (D) At constant rate

Q9. When an insulating medium is placed between two charges, the Coulomb's force:

(A) Increases (B) Decreases

(C) Becomes double (D) Remains same

Q10. The half life of radon gas is:

(A) 1620 years (B) 23.5 minutes

(C) 3.8 days (D) 4.5×10^9 years

Q11. Which basic force of nature has only repulsive nature?

(A) Weak nuclear force (B) Strong nuclear force

(C) Gravitational force (D) Electric force

Q12. The radius of 3rd orbit in hydrogen atom is:

(A) 0.477 nm (B) 0.053 nm

(C) 0.212 nm (D) 0.159 nm

Q13. A black body is:

(A) Ideal radiator (B) Ideal reflector

(C) Poor absorber (D) Poor radiator

Q14. Position was discovered by:

(A) De Broglie (B) Heisenberg

(C) Compton (D) Anderson

Q15. The current flowing into the base of a transistor is 25 microA while its collector current is 5mA. The current gain of transistor will be:

(A) 2000 (B) 200

(C) 500 (D) 1000

Q16. The Boolean expression of OR gate is:

(A) X = A+B (B) X = A·B

(C) X = A+B (negation over A+B) (D) X = A·B (negation over A·B)

Q17. An example of donor impurity is:

(A) Silicon (B) Germanium

(C) Phosphorous (D) Aluminum

Physics (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8475

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) The potential is constant throughout a given region of space. Is the electrical field zero or non-zero in this region? Explain.

(ii) Electric lines of force never cross. Why?

(iii) Why Gauss's law is used in electrostatics? How shape of Gaussian surface is chosen?

(iv) What is meant by potential gradient? How it is mathematically related to electric field?

(v) How can a current loop be used to determine presence of magnetic field in a given region of space?

(vi) How can you use magnetic field to separate isotopes of chemical element?

(vii) What is the working principle of a galvanometer?

(viii) What is meant by synchronization in CRO? How it is achieved to make the pattern stationary on screen of CRO?

(ix) Why are heavy nuclei unstable?

(x) What factors make a fusion reaction difficult to achieve?

(xi) How a Wilson Cloud chamber is used to determine information about charge, mass and energy of a radiating particle?

(xii) How can we justify emission of electron from inside of a nucleus during beta decay?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Define source of current. Write any two sources of current.

(ii) Starting from left first colour band is Red, second band is of violet colour, third band is of orange colour and fourth band is of silver colour. Calculate resistance and tolerance.

(iii) Describe a circuit which will give continuously varying potential.

(iv) Define amplitude modulation (A.M). Draw waveform of amplitude modulated wave.

(v) How does doubling the frequency affect the reactance of a capacitor and an inductor?

(vi) Explain the conditions under which electromagnetic waves are produced from a source.

(vii) Differentiate between glossy solids and polymeric solids.

(viii) Define stress and strain. Show that units of modulus of elasticity and stress are the same.

(ix) Differentiate between conductors and insulators with the help of energy band theory.

(x) Write Boolean expression and truth table of exclusive OR gate.

(xi) What are the biasing requirements for the junction of a transistor for its normal operation? Also draw circuit diagram of a common emitter amplifier.

(xii) What is the net charge on N-type and P-type substance?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Show that ε and $\Delta\varphi/\Delta t$ have the same units.

(ii) What is hysteresis loss? Write methods used to decrease the hysteresis loss.

(iii) Define alternating current. What is the time period of alternating current?

(iv) Give the statement of special theory of relativity.

(v) What is Compton wavelength and give its numerical value?

(vi) When does light behave as a wave and when does it behave as a particle?

(vii) Will higher frequency light eject greater number of electrons than low frequency light?

(viii) What are the advantages of lasers over ordinary light?

(ix) Is energy is conserved when an atom emits a photon of light?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Physics (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8472

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. A particle carrying a charge of 2e falls through a potential difference of 10V. The energy acquired by it is:

(A) 2eV
(B) 5eV
(C) 10eV
(D) 20eV

Q2. In a capacitor energy is stored in:

(A) Electric field
(B) Magnetic field
(C) Gravitational field
(D) Nuclear field

Q3. The charge carriers in electrolyte are:

(A) Free electrons
(B) Positive and negative ions
(C) Free electrons and ions
(D) Electrons and holes

Q4. Output waveform of sweep or time base generator is:

(A) Saw tooth wave
(B) Sinusoidal wave
(C) Square wave
(D) Digital wave

Q5. An alpha particle of charge 2e enters a uniform magnetic field of 0.1T with velocity $10ms^{-1}$ perpendicularly, the magnetic force acting on it will be:

(A) $1.6 \times 10^{-19}N$
(B) $3.2 \times 10^{-19}N$
(C) $6.4 \times 10^{-19}N$
(D) Zero

Q6. Lenz's law is in accordance with the law of conservation of:

(A) Charge
(B) Mass
(C) Momentum
(D) Energy

Q7. Eddy currents are setup in a direction:

(A) Parallel to flux
(B) Antiparallel to flux
(C) Perpendicular to flux
(D) At an angle 45° to the flux

Q8. The unit of impedance is:

(A) Ohm
(B) Farad
(C) Volt
(D) $(Ohm)^{-1}$

Q9. The power factor in a series resonance circuit at resonance is:

(A) 0
(B) 1
(C) 1
(D) Infinity

Q10. The units of modulus of elasticity are the same as those of:

(A) Stress
(B) Strain
(C) Power
(D) Work done

Q11. In case of non-inverting operational amplifier, if $R_1 = R_2/2$, then:

(A) Vout = 2 Vin
(B) Vin = 2 Vout
(C) Vout = 3 Vin
(D) Vin = 3 Vout

Q12. $X = A\cdot\bar{B} + \bar{A}\cdot B$ is the mathematical mutation for:

(A) NOR gate
(B) NAND gate
(C) XOR gate
(D) XNOR gate

Q13. The slope of the maximum K.E of photoelectrons versus light frequency graph represents:

(A) Momentum
(B) Planck's constant
(C) Maximum wavelength
(D) Work function

Q14. The minimum energy required for pair production is:

(A) 0.51 MeV
(B) 1.51 MeV
(C) 1.02 MeV
(D) 0.051 MeV

Q15. The radius of second Bohr radius for hydrogen atom is:

(A) 0.053 nm
(B) 0.212 nm
(C) 0.106 nm
(D) 0.848 nm

Q16. The dead time for G.M counter is:

(A) 10^{-3}s
(B) 10^{-4}s
(C) 10^{-5}s
(D) 10^{-8}s

Q17. Unit of radioactivity is curie (Ci). Which is equal to ____ disintegration per second.

(A) 3.74×10^9 (B) 3.7×10^{10}

(C) 3.64×10^9 (D) 4.5×10^9

Physics (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1212-XII124

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Define dielectric coefficient of capacitance.

(ii) Show that Ohm × Farad = Second.

(iii) Electric lines of force never cross. Why?

(iv) In the presence of dielectric why potential difference decreases?

(v) Why does the picture on a T.V screen becomes distorted when a magnet is brought near the screen?

(vi) What is Lorentz force? Give the role of electric and magnetic force in this regard.

(vii) Do two long and parallel current carrying wires attract each other? Explain.

(viii) A power line 20m high carries a current 200 A. Find the magnetic field of the wire at the ground.

(ix) If decay constant of a radioactive isotope is 0.3465 hr^{-1}. What will be its half-life?

(x) What is nuclear transmutation? Give one example.

(xi) What is radiography? Write its one use.

(xii) Discuss the advantages and disadvantages of nuclear power compared to the use of fossil fuel generated power.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Write names of two devices in which resistance decreases due to increase in temperature.

(ii) Do bends in a wire affect its electrical resistance? Explain.

(iii) What are the difficulties in testing whether the filament of a lighted bulb obeys Ohm's law?

(iv) In RLC series circuit, the impedance of the circuit at resonance is resistive. Why?

(v) A sinusoidal current has rms value of 10A. What is the maximum or peak value?

(vi) Name the device that will (a) permit flow of direct current but oppose the flow of alternating current (b) permit flow of alternating current but not the direct current.

(vii) Differentiate between glossy solids and polymeric solids.

(viii) Define modulus of elasticity. Show that the units of modulus of elasticity and stress are the same.

(ix) What is meant by strain energy?

(x) Summarize the advantages of photo diode.

(xi) What is net charge on a n-type or a p-type substance?

(xii) Why charge carriers are not present in the depletion region?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Four unmarked wires emerge from a transformer. What steps would you take to determine the turns ratio?

(ii) How would you position a flat loop of wire in a changing magnetic field so that there is no emf induced in the loop?

(iii) Define motional emf. Write its expression.

(iv) What happens to total radiation from a black body if its absolute temperature is doubled?

(v) Is it possible to create a single electron from energy? Explain.

(vi) State the two postulates of special theory of relativity.

(vii) What is work function? Write its mathematical relation with threshold frequency.

(viii) Mention any four applications of LASER.

(ix) Can X-rays be reflected, refracted, diffracted and polarized just like any other waves? Explain.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.